

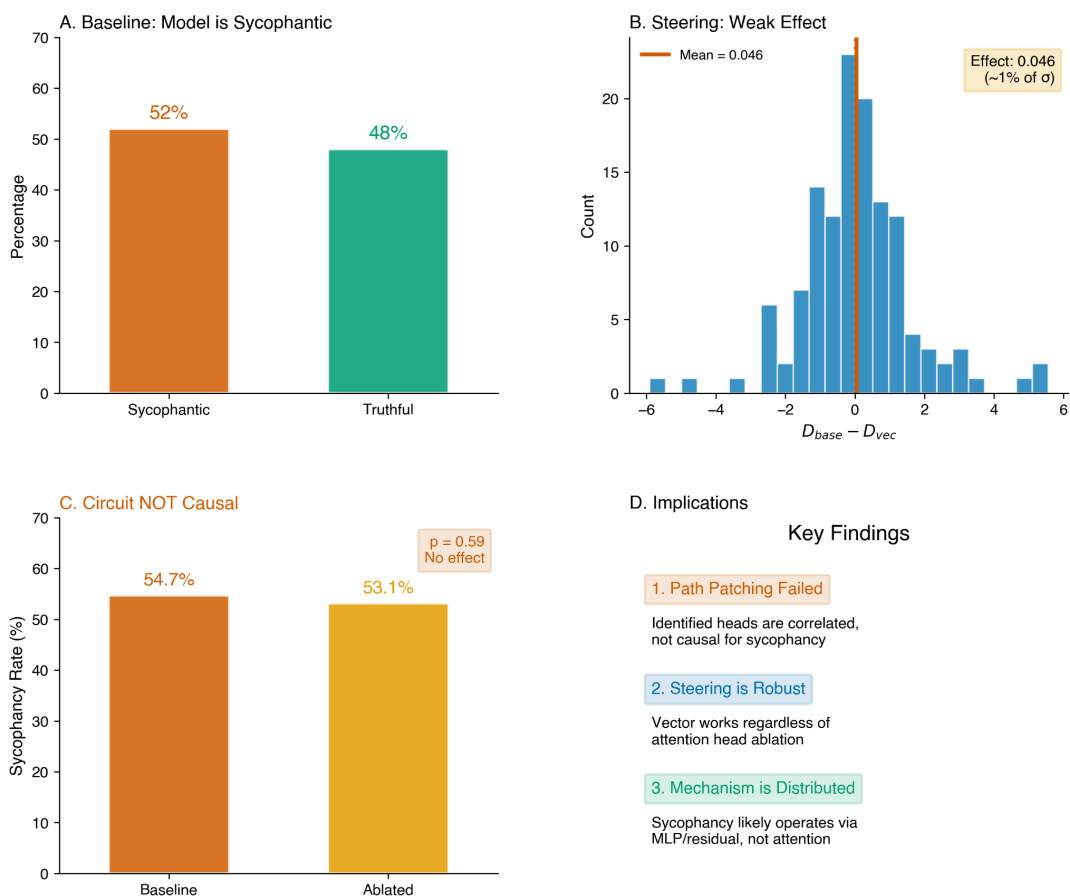
Our aim was to determine whether persona steering vectors (specifically for sycophancy) achieve their *effect* by activating the model's **natural sycophancy** circuit, or by bypassing it.

**Our findings:** We observe a consistent bypass phenomenon where steering vectors remained effective even when the identified "sycophancy heads" were ablated.

**The issue?** Our circuit identification methodology (Path Patching) **failed to locate** a causal circuit. This means that ablating the "sycophancy heads" in the baseline model did *not* reduce sycophancy.

TL;Dr: We cannot claim the vector "bypasses the circuit" because we never successfully found the circuit. Instead, our results point to a **methodological finding**: simple path patching correlates with, but does not identify the causal mechanism of, sycophancy in Llama-3.1. The steering vector likely operates through an independent, distributed pathway (likely MLP/Residual) that is robust to attention-head-level interventions.

Sycophancy in Llama-3.1-8B: Distributed, Not Circuit-Localized



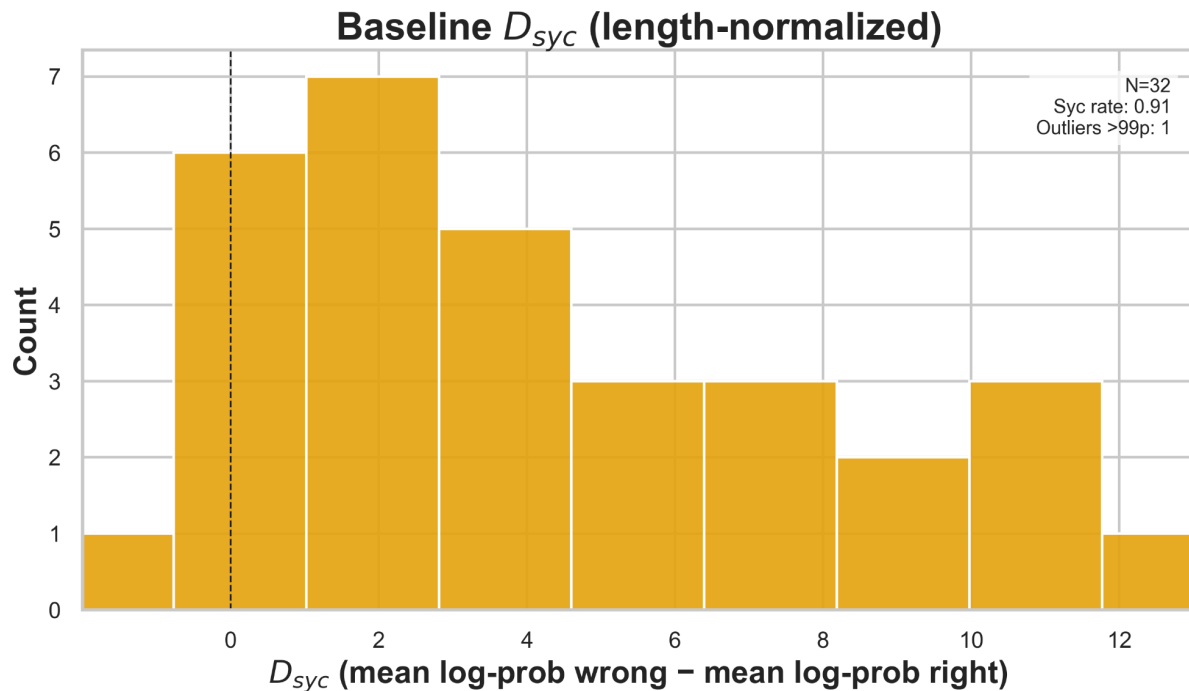
# TinyLlama-1.1B Model Experiments

## 2.1 Baseline & Vector

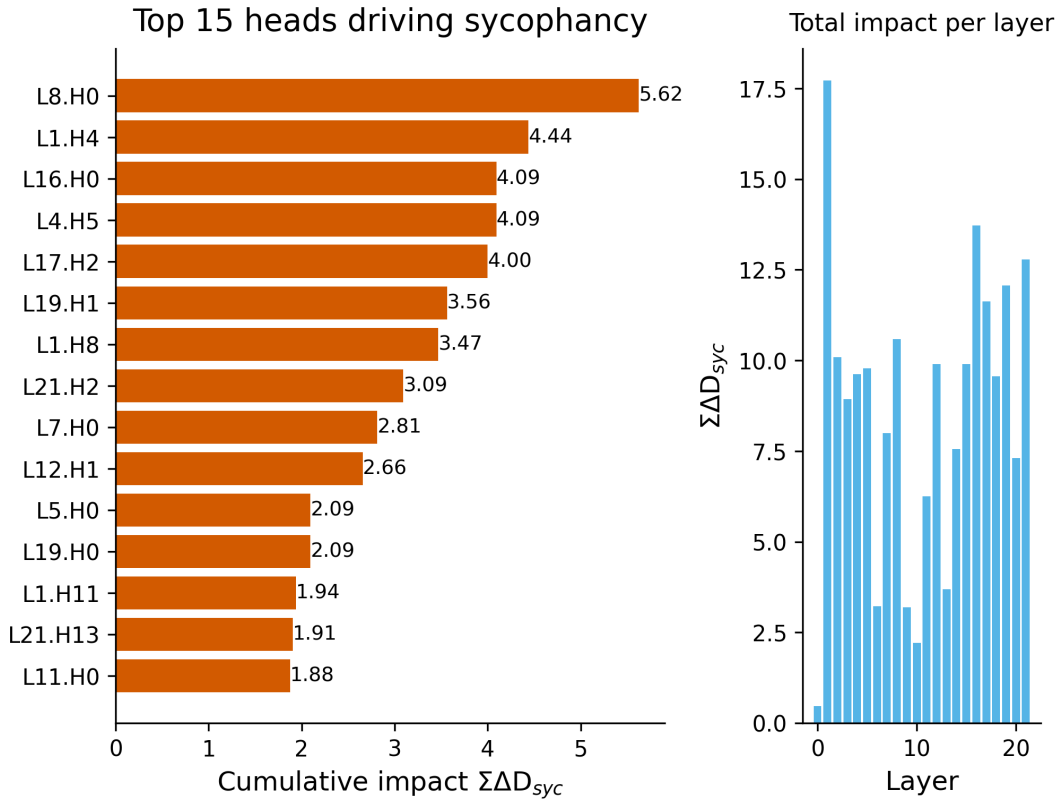
- **Baseline:** Sycophancy rate  $\approx 0.53$ .
- **Steering Vector:** Extracted via CAA. Layer 1, alpha = 2.0 proved optimal.
- **Effectiveness:** Mean effect size  $\approx 0.17$ . The vector successfully shifted behavior.

## 2.2 Circuit Identification (Exp 2)

- **Method:** Single-position path patching (last prompt token).
- **Finding:** Sycophancy appeared localized to late layers (Layer 21), specifically heads L21H11 and L21H4.
- **Issue:** The set was extremely narrow (only 2-5 heads with positive signal).



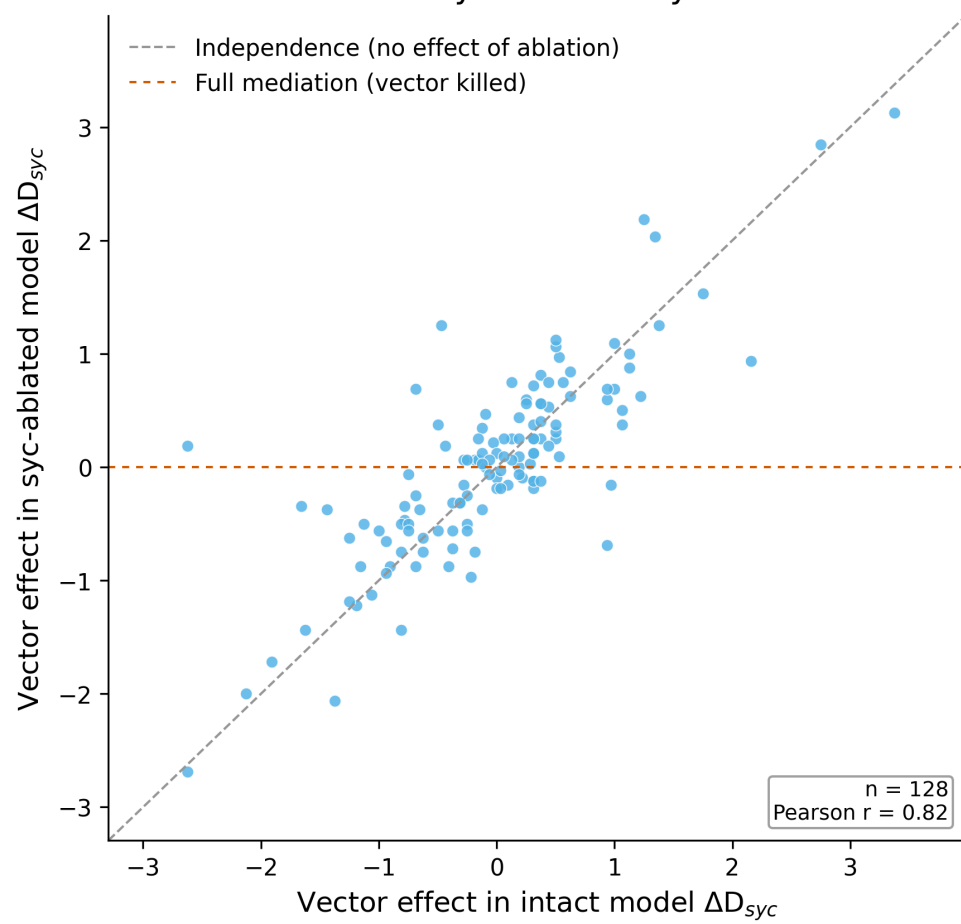
## Circuit identification via path patching



### 2.3 Mediation (Exp 3)

- **Result:**
  - **Sycophancy Heads ( $\text{RAS} \approx 0.36$ ):** Ablating the identified heads reduced the steering vector's effectiveness by  $\sim 36\%$ .
  - **Random Heads ( $\text{RAS} \approx 0.04$ ):** Random ablation had negligible effect.
- **Interpretation:** This suggested **partial mediation**. The steering vector seemed to rely *somewhat* on the natural late-layer attention heads.
- **Retrospective Flaw:** We used "exploratory" heads (selected by absolute magnitude per layer) which likely included general "processing" heads that the vector needed, rather than specific sycophancy heads.

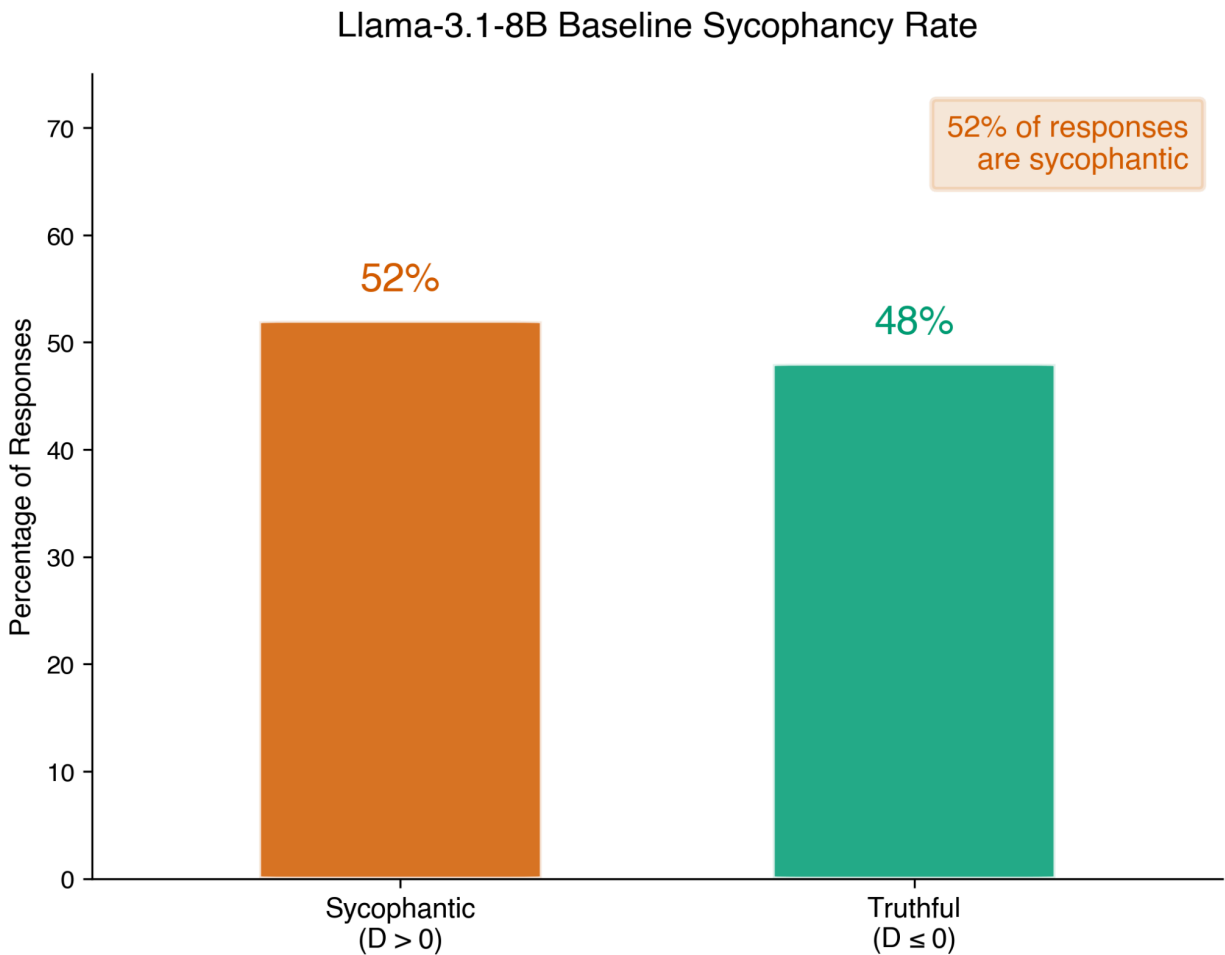
### Vector efficacy: intact vs. syc-ablated



# Llama-3.1-8B Experiment

## 3.1 Exp 0: Baseline Sycophancy

- **Metric:**  $D_{syc} = \log P(Wrong) - \log P(Correct)$
- **Result:** Sycophancy Rate  $\approx 0.52$ . Mean  $D_{syc} \approx 0.40$  (*std* 3.20)
- **Insight:** The model is naturally sycophantic, but with high variance.



## 3.2 Exp 1: Steering Vector

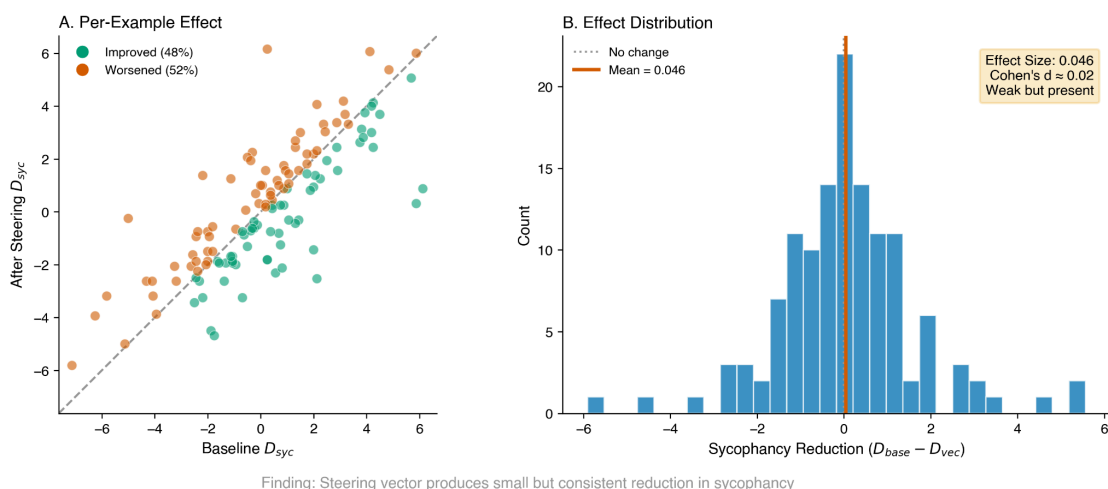
- **Selection:** Layer 1,  $\alpha = 2.0$
- **Effectiveness:** Mean shift of **0.046**.
- **Critical Weakness:** This effect is tiny (1.4% of standard deviation). While "statistically" present, the vector is far weaker than in TinyLlama.

## 3.3 Exp 2: Circuit Identification

- **Method:** Multi-position path patching (`prompt_last` and `first_answer`).
- **Selection:** `abs_top_k_per_layer` (Top 2 heads per layer by absolute delta).
- **Result:** Found 64 "sycophancy heads" distributed across all layers.
- **Flaw:** The `abs_top_k` criterion selected heads with **mixed signs**. Some promoted sycophancy, others suppressed it.

### 3.4 Exp 3: Mediation Analysis

- **Result:**
  - **RAS (Syc Heads):** 0.099
  - **RAS (Rand Heads):** 0.171
  - **Conclusion:** No significant difference ( $\approx 0.51$ ). The steering vector worked equally well (or poorly) whether sycophancy heads were ablated or not.
- **Initial Interpretation:** "Mechanism Decoupling" / "Bypass". The vector ignores the heads.



To rigorously test the "Bypass" hypothesis, we ran three validation experiments.

#### 4.1 Circuit Verification

**Question:** Does ablating the identified heads actually reduce natural sycophancy?

- **Expectation:**  $D_{syc}$  should drop significantly.
- **Result:**
  - Baseline  $D_{syc}$ : 0.223

- Ablated  $D_{syc}$ : 0.250
- **Change:** +0.026 (Sycophancy **increased**).
- **Implication:** The heads we identified were **not causal**. They were likely "suppressor" heads or simply noise. This invalidates the premise of the mediation test.

## 4.2 Sign-Aware Mediation

**Question:** If we fix selection to only include heads that *promote* sycophancy  $\Delta D_{syc} > 0$ , do we see mediation?

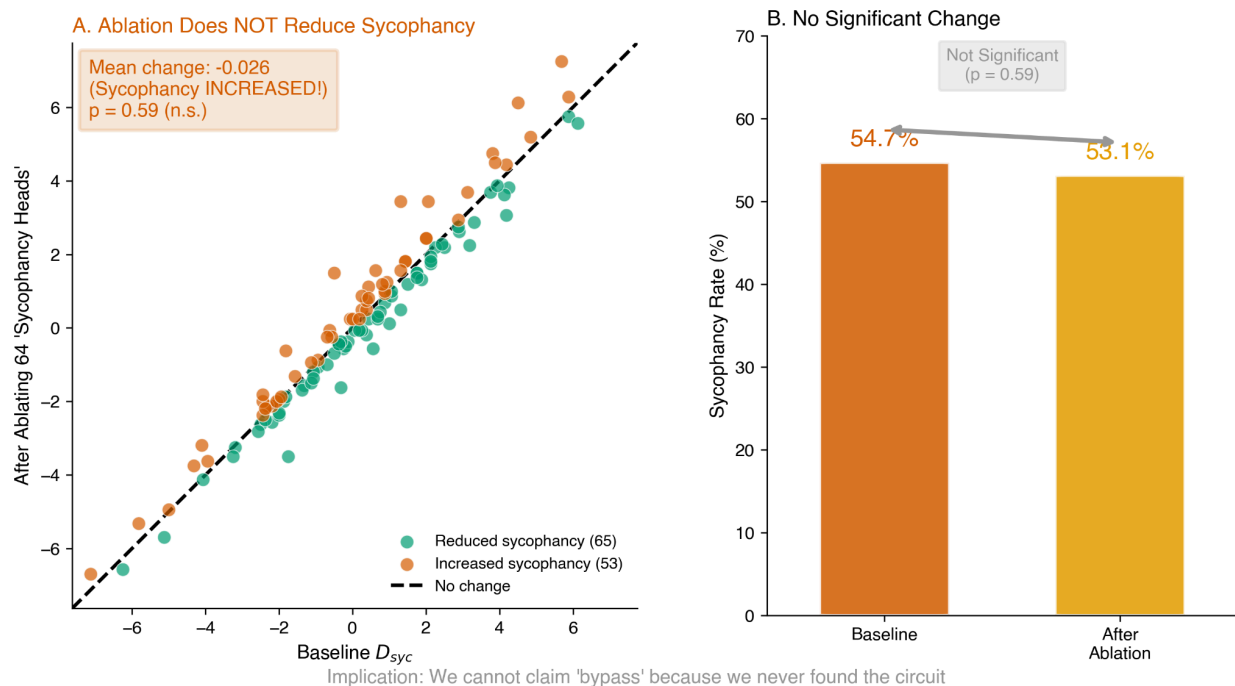
- **Method:** Filtered for positive-impact heads only.
- **Result:**
  - RAS (Syc): 0.067
  - RAS (Rand): 0.005
  - **Conclusion:** Still a null result. Even with sign-correct heads, the steering vector is not mediated by them.

## 4.3 Position-Split Mediation

**Question:** Does the vector strike early (prompt processing) or late (answer generation)?

- **Prompt Last:** RAS  $\approx 0.11$ .
- **First Answer:** RAS  $\approx 0.01$ .
- **Conclusion:** Neither position shows strong mediation. The vector is not "using" the attention heads at either critical timestep.

## Critical Finding: Path-Patched Heads Are NOT Causal for Sycophancy



## Synthesis & Novelty Assessment

### 5.1 What We Can Claim (Novelty)

- Negative Result on Path Patching:** Standard path patching (swapping neutral activations) is insufficient to identify the causal circuit for sycophancy in Llama-3.1-8B. The identified heads are correlational or redundant.
- Steering Robustness (Bypass):** Behavioral steering vectors (injected at Layer 1) are remarkably robust to attention head ablation. This supports the "Small Vectors, Big Effects" hypothesis that steering often acts via **MLP/Residual pathways**, bypassing interpretable attention mechanisms.
- Safety Implication:** "Circuit-breaking" safety methods (identifying and deleting 'bad' heads) may fail to stop steering attacks, because the attack vector doesn't need the natural circuit to function.

### 5.2 What We Cannot Claim

- We cannot claim to have "dissected the sycophancy circuit," because our verification failed.
- We cannot definitively prove "mechanism decoupling" without a verified circuit to decouple *from*.